Task 3 — Microservices with Hazelcast

GitHub: https://github.com/YOUR_USERNAME/YOUR_REPO/tree/micro_hazelcast

1. POST 10 transactions

Sending 10 transactions msg1-msg10 via facade-service

POST requests and responses:

```
[Facade] POST /log -> http://logging-service-1:8001
[Facade] POST tx=8996d761-2199-4ace-bff4-686ebd3e41dc user=user1 amount=+100.00 -> balance=100.00
INFO:     172.20.0.1:46132 - "POST /transaction HTTP/1.1" 201 Created
[Facade] POST /log -> http://logging-service-1:8001
[Facade] POST tx=66fe978b-9ca8-4504-b0de-14a2d8a5ccd8 user=user2 amount=+200.00 -> balance=200.00
INFO:     172.20.0.1:46132 - "POST /transaction HTTP/1.1" 201 Created
[Facade] POST /log -> http://logging-service-2:8001
[Facade] POST tx=75856a91-2b4a-4dea-9785-8aae9c4ca3e1 user=user3 amount=+300.00 -> balance=300.00
INFO:     172.20.0.1:46132 - "POST /transaction HTTP/1.1" 201 Created
[Facade] POST /log -> http://logging-service-2:8001
[Facade] POST tx=2e0eb301-b6d0-4f24-96a5-9aab6cb8b415 user=user4 amount=+400.00 -> balance=400.00
INFO:     172.20.0.1:46132 - "POST /transaction HTTP/1.1" 201 Created
[Facade] POST /log -> http://logging-service-1:8001
[Facade] POST tx=466f2955-6e42-4a72-ac31-7210f28346a8 user=user5 amount=+500.00 -> balance=500.00
INFO:     172.20.0.1:46132 - "POST /transaction HTTP/1.1" 201 Created
[Facade] POST /log -> http://logging-service-1:8001
[Facade] POST tx=5961c3fe-050b-43d1-b251-a8ab2e465f76 user=user6 amount=+600.00 -> balance=600.00
INFO:     172.20.0.1:46132 - "POST /transaction HTTP/1.1" 201 Created
[Facade] POST /log -> http://logging-service-2:8001
[Facade] POST tx=3e12edc0-53ff-4f48-bc5e-436ad0cde0f4 user=user7 amount=+700.00 -> balance=700.00
INFO:     172.20.0.1:46132 - "POST /transaction HTTP/1.1" 201 Created
[Facade] POST /log -> http://logging-service-3:8001
[Facade] POST tx=c7de0602-d29d-4d37-9dfe-6a29b5ccc0f5 user=user8 amount=+800.00 -> balance=800.00
INFO:     172.20.0.1:46132 - "POST /transaction HTTP/1.1" 201 Created
[Facade] POST /log -> http://logging-service-1:8001
[Facade] POST tx=c7e24770-7142-4231-9d26-8a99e4f65196 user=user9 amount=+900.00 -> balance=900.00
INFO:     172.20.0.1:46132 - "POST /transaction HTTP/1.1" 201 Created
[Facade] POST /log -> http://logging-service-3:8001
[Facade] POST tx=1ed9811c-1888-4c63-8f1c-d4e3a39e320b user=user10 amount=+1000.00 -> balance=1000.00
INFO:     172.20.0.1:46132 - "POST /transaction HTTP/1.1" 201 Created
```

2. Logging service distribution

logging-service-1 console:

```
[Logging-1] Connected to Hazelcast at hz1:5701
[Logging-1] Stored tx=8996d761-2199-4ace-bff4-686ebd3e41dc user=user1 amount=+100.00
INFO:     172.20.0.10:46416 - "POST /log HTTP/1.1" 201 Created
[Logging-1] Stored tx=66fe978b-9ca8-4504-b0de-14a2d8a5ccd8 user=user2 amount=+200.00
INFO:     172.20.0.10:46424 - "POST /log HTTP/1.1" 201 Created
[Logging-1] Stored tx=466f2955-6e42-4a72-ac31-7210f28346a8 user=user5 amount=+500.00
INFO:     172.20.0.10:46440 - "POST /log HTTP/1.1" 201 Created
[Logging-1] Stored tx=5961c3fe-050b-43d1-b251-a8ab2e465f76 user=user6 amount=+600.00
INFO:     172.20.0.10:46452 - "POST /log HTTP/1.1" 201 Created
[Logging-1] Stored tx=c7e24770-7142-4231-9d26-8a99e4f65196 user=user9 amount=+900.00
INFO:     172.20.0.10:46468 - "POST /log HTTP/1.1" 201 Created
```

logging-service-2 console:

```
[Logging-2] Connected to Hazelcast at hz2:5701
[Logging-2] Stored tx=75856a91-2b4a-4dea-9785-8aae9c4ca3e1 user=user3 amount=+300.00
INFO:     172.20.0.10:40234 - "POST /log HTTP/1.1" 201 Created
[Logging-2] Stored tx=2e0eb301-b6d0-4f24-96a5-9aab6cb8b415 user=user4 amount=+400.00
INFO:     172.20.0.10:40240 - "POST /log HTTP/1.1" 201 Created
[Logging-2] Stored tx=3e12edc0-53ff-4f48-bc5e-436ad0cde0f4 user=user7 amount=+700.00
INFO:     172.20.0.10:40254 - "POST /log HTTP/1.1" 201 Created
```

logging-service-3 console:

```
[Logging-3] Connected to Hazelcast at hz3:5701
[Logging-3] Stored tx=c7de0602-d29d-4d37-9dfe-6a29b5ccc0f5 user=user8 amount=+800.00
INFO:     172.20.0.10:34850 - "POST /log HTTP/1.1" 201 Created
[Logging-3] Stored tx=1ed9811c-1888-4c63-8f1c-d4e3a39e320b user=user10 amount=+1000.00
INFO:     172.20.0.10:34854 - "POST /log HTTP/1.1" 201 Created
```

3. GET transactions

Reading transactions and balances via facade-service

```
INFO:     172.20.0.1:40320 - "GET /accounts HTTP/1.1" 200 OK
[Facade] GET /logs/user1 -> http://logging-service-2:8001
INFO:     172.20.0.1:40320 - "GET /user/user1 HTTP/1.1" 200 OK
```

4. Fault Tolerance

4.1 Kill one logging-service instance

4.2 Kill two logging-service instances

POST and GET with logging-service-1 down:

```
>> Invoke-RestMethod -Method POST -Uri "http://localhost:8000/transaction" -ContentType "application/json" -Body '{"user_id": "test", "amount": 111}'
>> Invoke-RestMethod -Uri "http://localhost:8000/accounts"
logging-service-1

transaction_id                         balance
--------------                         -------
ea05c04b-34d6-4d71-a517-44963dd3bd73   111.0
```

We see that it still works correctly

POST and GET with logging-service-1 and logging-service-2 down:

```
PS D:\UNI_UCU\APZ\apz_homework_2> docker stop logging-service-2
>> Invoke-RestMethod -Method POST -Uri "http://localhost:8000/transaction" -ContentType "application/json" -Body '{"user_id": "test2", "amount": 222}'
>> Invoke-RestMethod -Uri "http://localhost:8000/accounts"
logging-service-2

transaction_id                         balance
--------------                         -------
625213f4-53c9-4300-8eb9-2ef3b3edeadc   222.0
```

4.3 Kill one Hazelcast node

POST and GET with hz1 down:

```
PS D:\UNI_UCU\APZ\apz_homework_2> docker stop hz1
>> Invoke-RestMethod -Method POST -Uri "http://localhost:8000/transaction" -ContentType "application/json" -Body '{"user_id": "test3", "amount": 333}'
>> Invoke-RestMethod -Uri "http://localhost:8000/accounts"
hz1

transaction_id                         balance
--------------                         -------
cd6457c7-b70f-4c24-b45b-73635ed960be   333.0
```

4.4 Kill two Hazelcast nodes

POST and GET with hz1 and hz2 down:

```
PS D:\UNI_UCU\APZ\apz_homework_2> docker stop hz2
>> Invoke-RestMethod -Method POST -Uri "http://localhost:8000/transaction" -ContentType "application/json" -Body '{"user_id": "test4", "amount": 444}'
>> Invoke-RestMethod -Uri "http://localhost:8000/accounts"
hz2
Invoke-RestMethod : {"detail":"All logging service instances unavailable"}
At line:2 char:1
+ Invoke-RestMethod -Method POST -Uri "http://localhost:8000/transactio ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-RestMethod], WebException
    + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeRestMethodCommand

balances
--------
@{test4=444.0}
```

4.5 Fault tolerance analysis

- 1 logging service down: facade falls back to remaining instances automatically
- 2 logging services down: facade falls back to the last remaining instance
- 1 Hazelcast node down: remaining nodes hold all data, system works
- 2 Hazelcast nodes down: all logging services lose connection (smart_routing=False means each instance only knows its assigned node), POST fails with 'All logging service instances unavailable', GET still works since MongoDB is unaffected

5. Performance Testing

Comparing results with Task 1 (plain in-memory storage)

Scenario 1 — 10 clients x 10K transactions on own accounts:

scenario1

Scenario 2 — 10 clients x 10K transactions on shared account:

scenario2

Performance analysis

PASTE YOUR ANALYSIS HERE:

- Task 1 results: req/s, logging avg ms, counter avg ms
- Task 3 results: req/s, logging avg ms, counter avg ms
- Why is Task 3 slower/faster?
- Hazelcast adds network overhead for logging (distributed map vs local dict)
- MongoDB adds persistence overhead for counter (disk vs in-memory)